NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

Team: CSE 4104-7A-02

Project Proposal

Project Title: AI Sales Analytics Dashboard

Course Name: Software Development III

Course Code: CSE 4104

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Introduction

The rapid proliferation of digital commerce and enterprise resource systems has resulted in businesses accumulating vast repositories of sales data that remain largely underutilised (Bello, Idemudia and Iyelolu, 2024). Despite this abundance of data, a significant proportion of small and medium enterprises continue to rely on manual reporting methods that are neither scalable nor analytically capable. Artificial intelligence and machine learning have fundamentally changed how organisations can interact with their own data, enabling real-time forecasting, anomaly detection, and natural language querying that were previously accessible only to large corporations with dedicated data science teams. This project proposes the development of an AI Sales Analytics Dashboard — a full-stack, production-ready web application that integrates intelligent features directly into a business reporting interface. The system is designed to serve non-technical business users by making complex data analysis intuitive, automated, and immediately actionable.

Problem Statement and Objectives

Modern businesses generate enormous volumes of sales data daily, yet most small and medium enterprises lack the tools to extract actionable intelligence from it. Existing solutions such as spreadsheets or basic reporting tools require manual interpretation, which is slow, error-prone, and incapable of identifying hidden patterns. Business managers often make strategic decisions based on outdated or incomplete information, leading to missed opportunities, poor inventory control, and ineffective regional targeting. Sales anomalies — sudden drops or unexpected spikes — frequently go undetected until significant financial damage has already occurred (Choi et al., 2021). Furthermore, generating monthly sales reports manually consumes considerable time that could be redirected toward strategic planning. There is a clear and measurable gap between the volume of data businesses collect and their actual capacity to use it intelligently.

Objectives

This project aims to design and develop a full-stack, AI-integrated sales analytics dashboard that transforms raw sales data into intelligent, real-time business insights. The specific objectives are:

- To build a secure, role-based authentication system supporting Admin, Manager, and Viewer access levels.

- To develop a dynamic dashboard displaying revenue trends, product performance, and regional sales through interactive visual charts.
- To integrate AI-powered features including sales forecasting, anomaly detection, insight generation, and a natural language chatbot.
- To implement bulk CSV data upload, manual data entry, and automated AI-generated monthly report generation.
- To deploy the complete system as a live, production-ready web application with full documentation.

Features

The system is structured into five functional areas that together form a comprehensive business intelligence platform.

Module	Key Features
Authentication	Register, login, JWT authentication, role-based access (Admin, Manager, Viewer)
Sales Management	Add/edit/delete records, CSV bulk upload, date & region filtering
Dashboard & Analytics	Revenue line chart, product bar chart, regional pie chart, KPI summary cards
AI Features	Insight generator, sales forecaster, anomaly detector, chatbot, recommendation engine
Reports & Export	AI-written monthly report, PDF export, CSV export, low-performance alerts

Table 1: Core Feature Breakdown by Module

Table 1 outlines the five core modules, demonstrating that each layer of the system serves a distinct and non-redundant functional purpose. The dashboard module will render real-time visualisations of key performance indicators including total revenue, average order value, and monthly growth rate. The AI features module carries the highest academic weight in this assessment and is therefore the most technically developed component. Sales forecasting will be implemented using scikit-learn regression models trained on historical sales data, providing next-month revenue predictions with measurable accuracy (Zhao, Wang and Liu, 2020). The anomaly detection component will flag statistically unusual patterns in time-series sales data, alerting managers before losses escalate. The AI chatbot, powered by the OpenAI or Gemini API, will allow users to query sales data using natural language, such as asking which product performed best in a specific region during a given month.

Technology Stack

The technology stack has been selected for industry relevance, scalability, and suitability to the analytical nature of a sales intelligence platform.

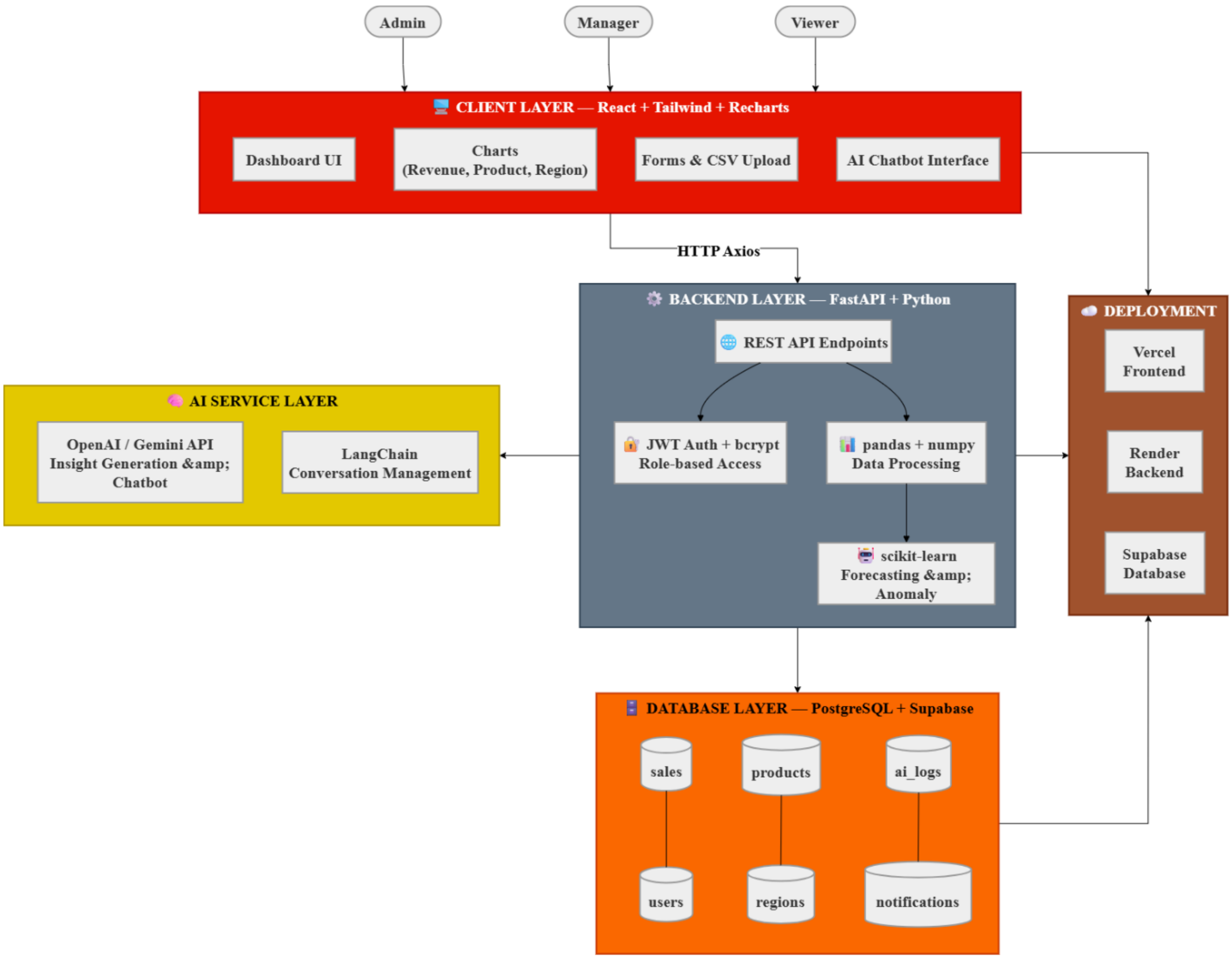

Figure 1: System Architecture Diagram

Python was selected as the backend language because sales analytics demands real-time data processing, statistical modelling, and machine learning integration — capabilities where Python's pandas, numpy, and scikit-learn libraries are industry standard (Rao and Narsimha, 2024). FastAPI was chosen over alternatives because it provides automatic Swagger API documentation, asynchronous request handling, and a significantly faster development cycle. The frontend will be built in React.js with Tailwind CSS for styling and Recharts for interactive data visualisation, ensuring a responsive interface across desktop, tablet, and mobile devices. PostgreSQL hosted on Supabase will serve as the relational database, managing structured sales records, user roles, product data, and AI query logs. LangChain will manage multi-turn AI conversations, enabling the chatbot to retain context across a session and deliver coherent, data-aware responses.

Expected Outcome

Upon completion, this system will deliver a fully deployed, AI-integrated sales analytics platform accessible via a live public URL, supported by complete technical documentation, a user manual, and a tested REST API. Business users will gain the ability to upload sales data, receive instant AI-generated insights, forecast next-period revenue, and detect anomalies without requiring any data science expertise. The system directly addresses the decision-making gap identified in the problem statement by transforming passive data storage into active, intelligent analysis (Prawira et al., 2025). Each team member will contribute a well-defined, independently verifiable module — frontend, backend, database, and AI integration — ensuring balanced workload distribution and individual accountability during the final viva evaluation.

Conclusion

This proposal presents a well-scoped, technically grounded, and academically justified plan for an AI Sales Analytics Dashboard that addresses a genuine problem faced by businesses lacking intelligent data analysis tools. The selected technology stack is industry-relevant, the feature set is realistic within the semester timeline, and the AI integration is substantive rather than superficial. The system architecture clearly separates concerns across four layers, supporting independent development by each team member while ensuring full integration at deployment. This project meets all mandatory requirements outlined in the CSE 4104 course handbook, including frontend, backend, database, authentication, AI integration, deployment, and documentation, and is designed to produce a portfolio-quality product that demonstrates real-world software engineering competence.

References

Bello, I.O., Idemudia, C. and Iyelolu, T.V. 2024. 'Scaling AI-driven sales analytics for predicting consumer behaviour and enhancing data-driven business decisions', *International Journal of Advanced Multidisciplinary Research and Studies*, vol. 4, no. 6, pp. 2181–2201. Available at: https://www.researchgate.net/publication/392063730

Udokwu, C., Brandtner, P., Darbanian, F. and Falatouri, T. 2023. 'Applications of large language models in business analytics: exemplary use cases in data preparation tasks', in *HCI International 2023 – Late Breaking Papers*, Lecture Notes in Computer Science, vol. 14059. Springer, Cham. DOI: https://doi.org/10.1007/978-3-031-48057-7_12

Rao, V.S. and Narsimha, G. 2024. 'Predictive analytics for business decision-making: unleashing the power of data-driven insights', in *Proceedings of the 2024 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*. IEEE. DOI: https://doi.org/10.1109/ICICT60155.2024.10724574

Prawira, M. et al. 2025. 'Exploring the role of machine learning and big data analytics in enhancing decision-making processes: a systematic literature review', *JOIV: International Journal on Informatics Visualization*, vol. 9, no. 4. DOI: https://doi.org/10.62527/joiv.9.4.3244

Ahmadov, Y. and Helo, P. 2023. 'Deep learning-based approach for forecasting intermittent online sales', *Discover Artificial Intelligence*, vol. 3, article 45. DOI: https://doi.org/10.1007/s44163-023-00085-1

Zhao, M., Wang, J. and Liu, G. 2020. 'Product sales forecasting based on improved LSTM neural network algorithm', *IEEE Access*, vol. 8, pp. 167901–167911. DOI: https://doi.org/10.1109/ACCESS.2020.3024795

Choi, K., Yi, J., Park, C. and Yoon, S. 2021. 'Deep learning for anomaly detection in time-series data: review, analysis, and guidelines', *IEEE Access*, vol. 9, pp. 120043–120065. DOI: https://doi.org/10.1109/ACCESS.2021.3107975

Zong, Z. and Guan, Y. 2024. 'AI-driven intelligent data analytics and predictive analysis in industry 4.0: transforming knowledge, innovation, and efficiency', *Journal of the Knowledge Economy*. DOI: https://doi.org/10.1007/s13132-024-02066-4